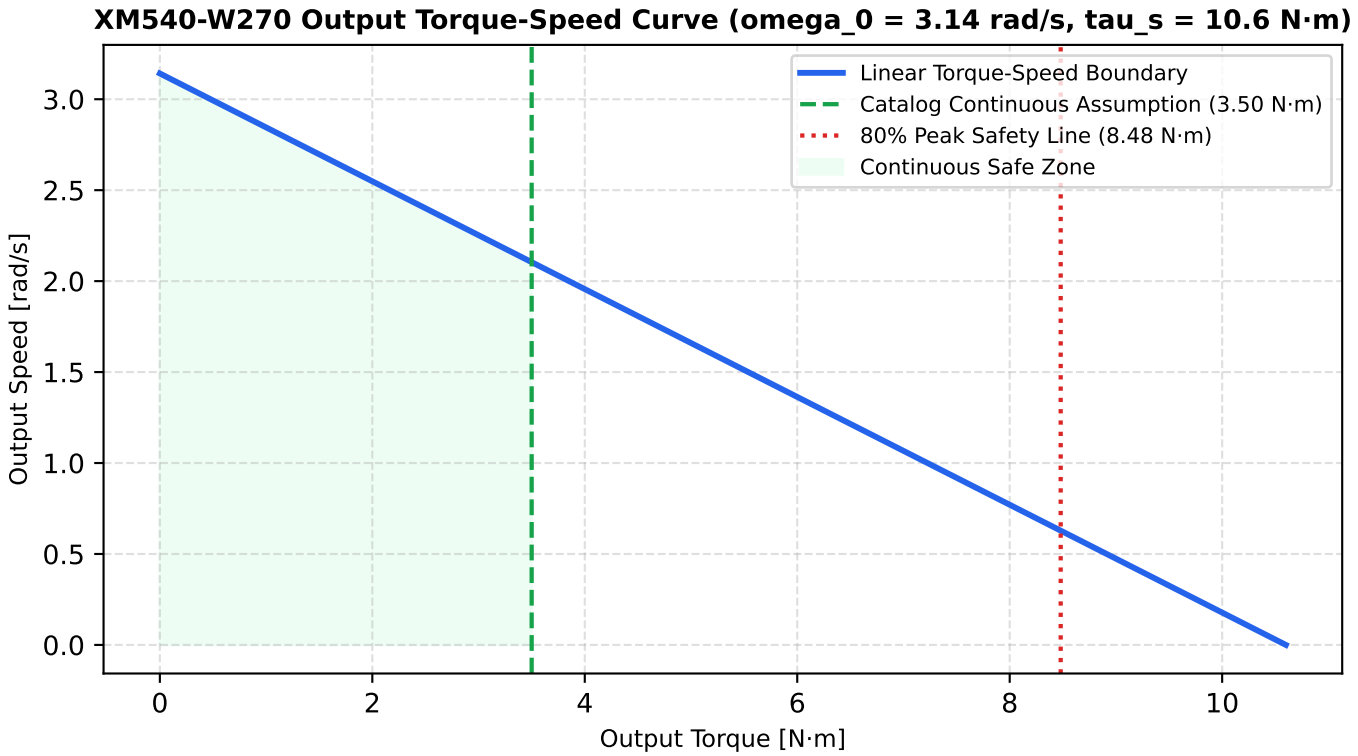
Part Number: XM540-W270

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	XM540-W270	DYNAMIXEL-X
Manufacturer	Vendor	ROBOTIS	Coreless DC Motor
Rated Input Voltage	V_rated	12.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	10.6 N.m (at 12.0V, 4.4A)	At 12.0 V
Continuous Rated Torque	tau_cont	3.50 N.m continuous	25 degC ambient, continuous duty
No-Load Speed	omega_0	30 rpm (at 12.0V)	Unloaded output shaft
Actuator Mass (Weight)	m_act	165.0 g	Includes housing & internal gearhead
Internal Reduction	i_int	272.5 : 1	Integrated gear stage
Envelope Dimensions	W x H x D	33.5 x 58.5 x 44.0 mm	Outer housing envelope



XM540-W270 — Project Reference & Review Notes

2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 33.5 mm x 58.5 mm x 44.0 mm
   - Catalog Mass (unverified): 165.0 g (0.1650 kg)
   - Mounting Pattern: Not verified; obtain manufacturer flange/shaft drawings.

3. Thermal & Duty-Cycle Notes
   - Catalog continuous torque is an input assumption until its manufacturer rating is check
   - The 0.33x stall estimate, where used, is a heuristic rather than a measured thermal rat
   - The linear torque-speed envelope is a screening model, not a measured performance curve

4. Project Reference Index
   - Part ID: XM540-W270 | Source Document: xm540-w270.pdf | Spec Table: Page 1

MotionFit-generated reference — not a manufacturer original; verify specifications